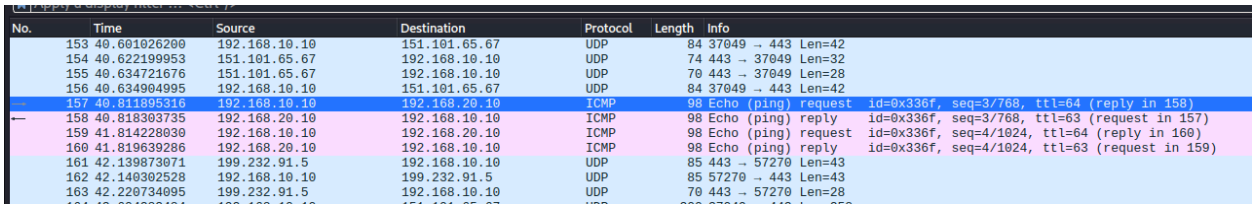
Home Lab Project Week 11, Day 1
Task:
Re-run selected reconnaissance and port-scanning activities from the previous testing phases and capture the traffic with Wireshark. Compare the results before and after firewall implementation.

Phase 1: Preparation & Wireshark Capture

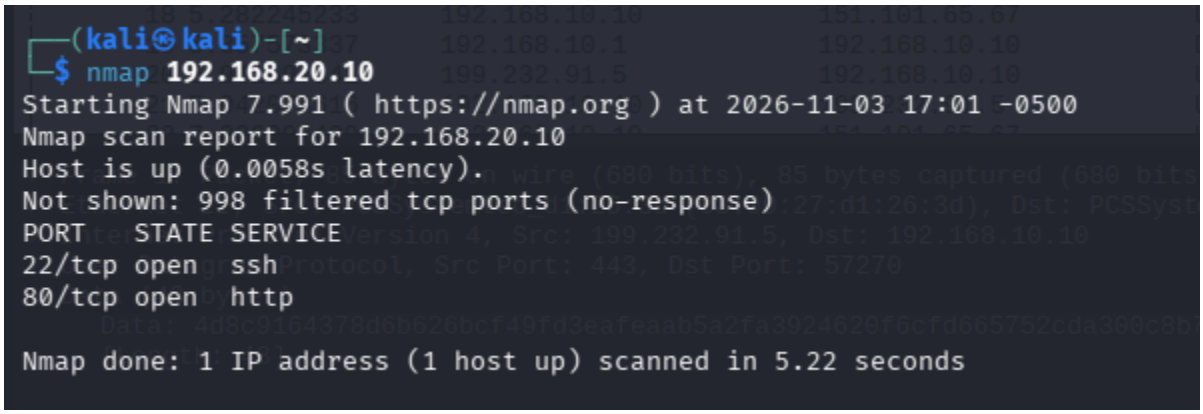
To get clean comparison data, you will run two distinct testing phases.

Test 1: Pre-Firewall (Baseline)

1. **Disable or bypass** your newly implemented firewall rules temporarily (or target a host/subnet where the rules are not yet applied).
2. Start a **Wireshark capture** on the target interface or a mirrored port.
3. Execute your selected reconnaissance scans (e.g., Nmap, Netcat).
4. Stop the capture and save it as `baseline_recon.pcapng`.

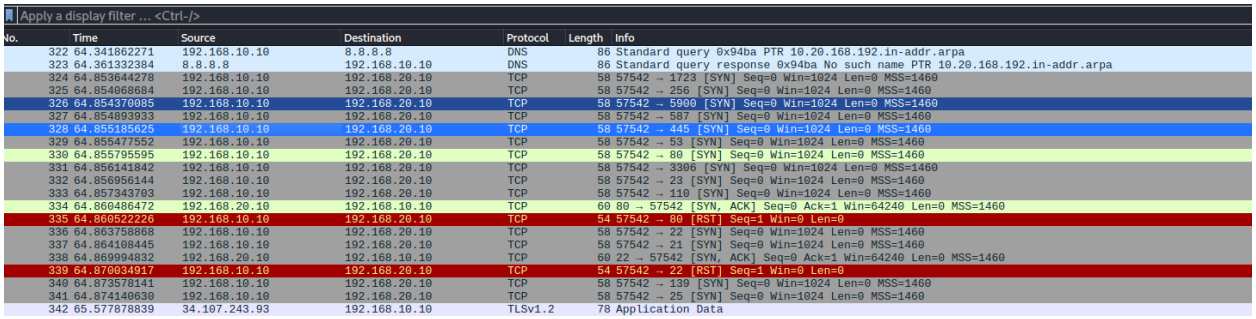


No.	Time	Source	Destination	Protocol	Length	Info
153	40.601026200	192.168.10.10	151.101.65.67	UDP	84	37049 → 443 Len=42
154	40.622199953	151.101.65.67	192.168.10.10	UDP	74	443 → 37049 Len=32
155	40.634721676	151.101.65.67	192.168.10.10	UDP	70	443 → 37049 Len=28
156	40.634904995	192.168.10.10	151.101.65.67	UDP	84	37049 → 443 Len=42
→ 157	40.811895316	192.168.10.10	192.168.20.10	ICMP	98	Echo (ping) request id=0x336f, seq=3/768, ttl=64 (reply in 158)
← 158	40.818303735	192.168.20.10	192.168.10.10	ICMP	98	Echo (ping) reply id=0x336f, seq=3/768, ttl=63 (request in 157)
159	41.814228030	192.168.10.10	192.168.20.10	ICMP	98	Echo (ping) request id=0x336f, seq=4/1024, ttl=64 (reply in 160)
160	41.819639286	192.168.20.10	192.168.10.10	ICMP	98	Echo (ping) reply id=0x336f, seq=4/1024, ttl=63 (request in 159)
161	42.139873071	199.232.91.5	192.168.10.10	UDP	85	443 → 57270 Len=43
162	42.140302528	192.168.10.10	199.232.91.5	UDP	85	57270 → 443 Len=43
163	42.220734095	199.232.91.5	192.168.10.10	UDP	70	443 → 57270 Len=28



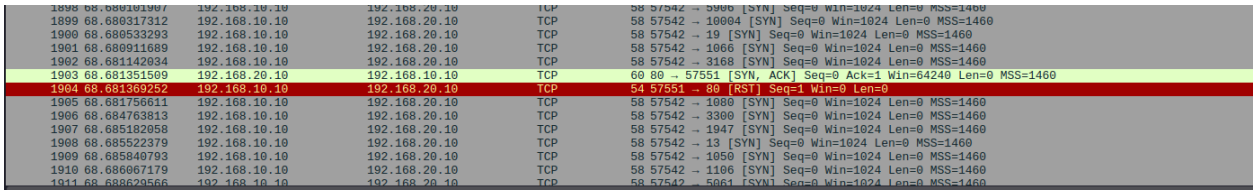
```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.20.10
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-03 17:01 -0500
Nmap scan report for 192.168.20.10
Host is up (0.0058s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 5.22 seconds
```



Apply a display filter ... <Ctrl-/>

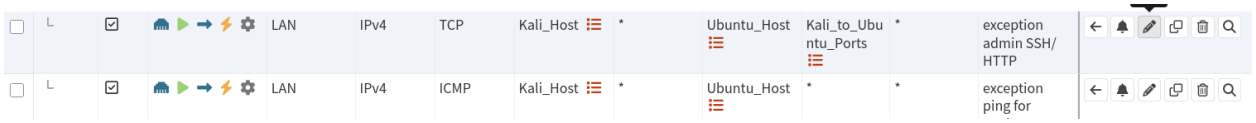
No.	Time	Source	Destination	Protocol	Length	Info
322	64.341862271	192.168.10.10	8.8.8.8	DNS	86	Standard query 0x94ba PTR 10.20.168.192.in-addr.arpa
323	64.361332384	8.8.8.8	192.168.10.10	DNS	86	Standard query response 0x94ba No such name PTR 10.20.168.192.in-addr.arpa
324	64.853644278	192.168.10.10	192.168.20.10	TCP	58	57542 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
325	64.854068684	192.168.10.10	192.168.20.10	TCP	58	57542 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
326	64.854370085	192.168.10.10	192.168.20.10	TCP	58	57542 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
327	64.854893933	192.168.10.10	192.168.20.10	TCP	58	57542 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
328	64.855185625	192.168.10.10	192.168.20.10	TCP	58	57542 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
329	64.855477552	192.168.10.10	192.168.20.10	TCP	58	57542 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
330	64.855795595	192.168.10.10	192.168.20.10	TCP	58	57542 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
331	64.856141842	192.168.10.10	192.168.20.10	TCP	58	57542 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
332	64.856956144	192.168.10.10	192.168.20.10	TCP	58	57542 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
333	64.857343703	192.168.10.10	192.168.20.10	TCP	58	57542 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
334	64.860486472	192.168.20.10	192.168.10.10	TCP	60	80 → 57542 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
335	64.860522226	192.168.10.10	192.168.20.10	TCP	54	57542 → 80 [RST] Seq=1 Win=0 Len=0
336	64.863758868	192.168.10.10	192.168.20.10	TCP	58	57542 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
337	64.864108445	192.168.10.10	192.168.20.10	TCP	58	57542 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
338	64.869994832	192.168.20.10	192.168.10.10	TCP	60	22 → 57542 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
339	64.870034917	192.168.10.10	192.168.20.10	TCP	54	57542 → 22 [RST] Seq=1 Win=0 Len=0
340	64.873578141	192.168.10.10	192.168.20.10	TCP	58	57542 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
341	64.874140630	192.168.10.10	192.168.20.10	TCP	58	57542 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
342	65.577878839	34.107.243.93	192.168.10.10	TLSv1.2	78	Application Data



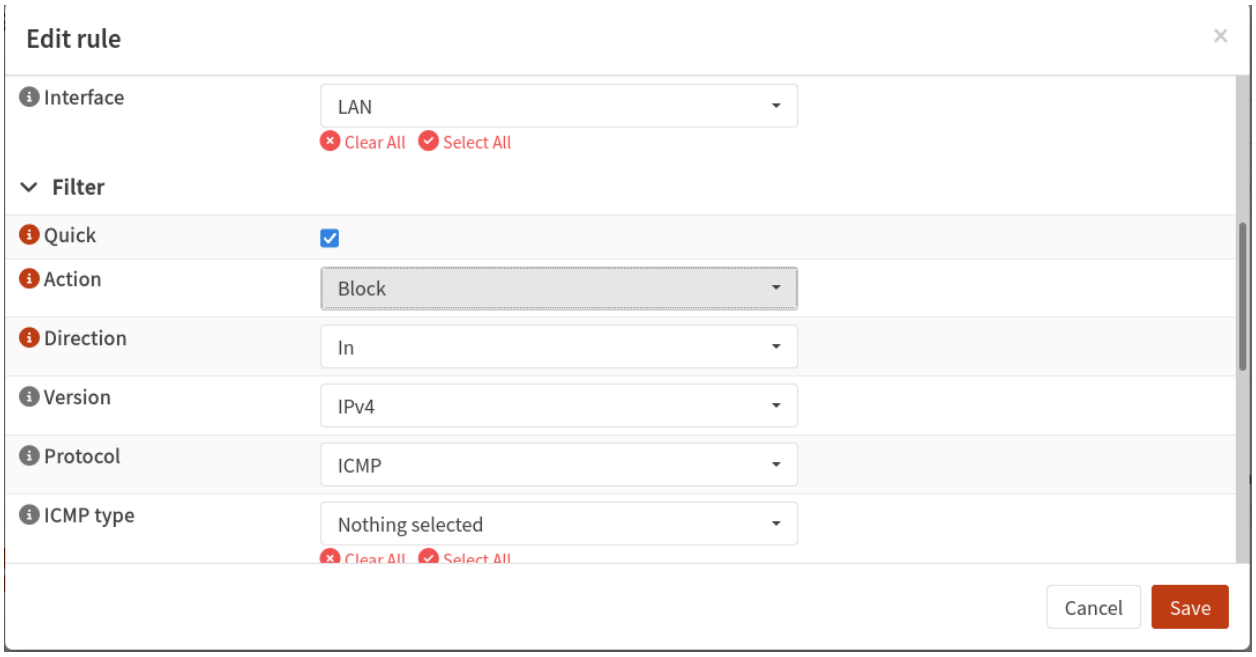
1898	68.680101907	192.168.10.10	192.168.20.10	TCP	58	57542 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1899	68.680317312	192.168.10.10	192.168.20.10	TCP	58	57542 → 10004 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1900	68.680533293	192.168.10.10	192.168.20.10	TCP	58	57542 → 19 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1901	68.680911689	192.168.10.10	192.168.20.10	TCP	58	57542 → 1066 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1902	68.681142034	192.168.10.10	192.168.20.10	TCP	58	57542 → 3169 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1903	68.681351509	192.168.20.10	192.168.10.10	TCP	60	80 → 57551 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
1904	68.681369252	192.168.10.10	192.168.20.10	TCP	54	57551 → 80 [RST] Seq=1 Win=0 Len=0
1905	68.681756611	192.168.10.10	192.168.20.10	TCP	58	57542 → 1080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1906	68.684763813	192.168.10.10	192.168.20.10	TCP	58	57542 → 3300 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1907	68.685182058	192.168.10.10	192.168.20.10	TCP	58	57542 → 1947 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1908	68.685522379	192.168.10.10	192.168.20.10	TCP	58	57542 → 13 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1909	68.685840793	192.168.10.10	192.168.20.10	TCP	58	57542 → 1050 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1910	68.686067179	192.168.10.10	192.168.20.10	TCP	58	57542 → 1106 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
1911	68.688629566	192.168.10.10	192.168.20.10	TCP	58	57542 → 5061 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Test 2: Post-Firewall (Hardened)

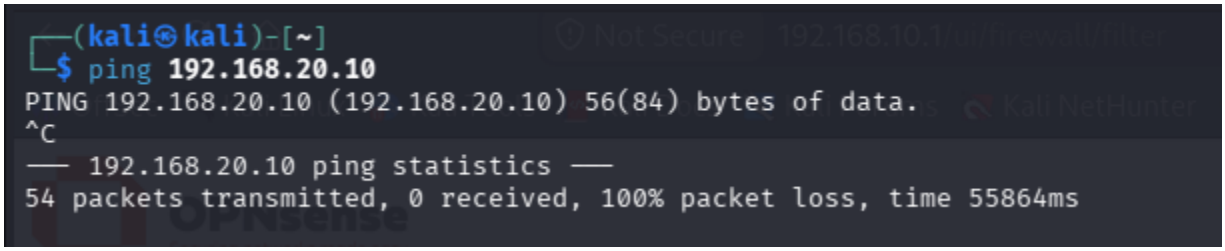
1. **Enable** your firewall rules (e.g., blocking unauthorized ICMP, dropping stealth scans, or limiting connection rates).
2. Start a **new Wireshark capture** on the same target interface.
3. Execute the exact same reconnaissance scans from the same attacker IP.
4. Stop the capture and save it as `hardened_recon.pcapng`.



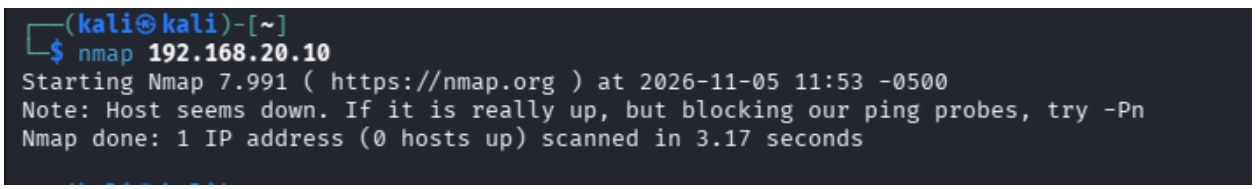
			LAN	IPv4	TCP	Kali_Host	*	Ubuntu_Host	Kali_to_Ubuntu_Ports	*	exception admin SSH/ HTTP
			LAN	IPv4	ICMP	Kali_Host	*	Ubuntu_Host	*	*	exception ping for



I managed to block the ping from Kali

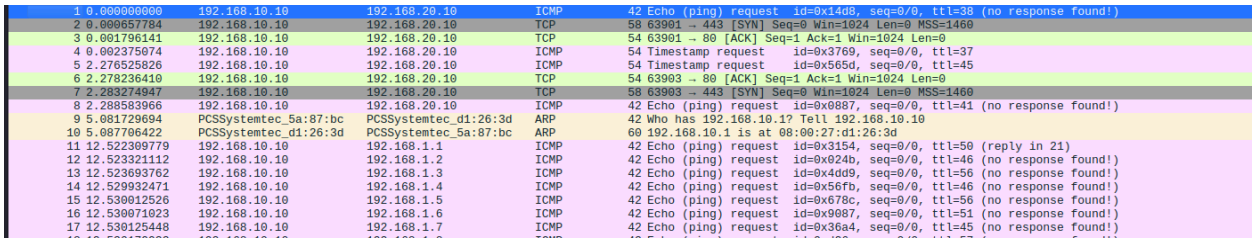


And the nmap too

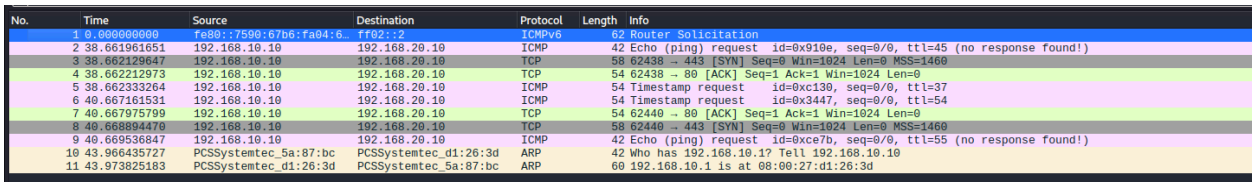


```
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.1.0/24
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-05 11:53 -0500
Nmap scan report for 192.168.1.1
Host is up (0.039s latency).
Nmap scan report for 192.168.1.186
Host is up (0.100s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 9.08 seconds
```

Capture on the wireshark



1	0.000000000	192.168.10.10	192.168.20.10	ICMP	42 Echo (ping) request id=0x14d8, seq=0/0, ttl=38 (no response found!)
2	0.000657784	192.168.10.10	192.168.20.10	TCP	58 63901 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
3	0.001796141	192.168.10.10	192.168.20.10	TCP	54 63901 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
4	0.002375074	192.168.10.10	192.168.20.10	ICMP	54 Timestamp request id=0x3769, seq=0/0, ttl=37
5	2.276525826	192.168.10.10	192.168.20.10	ICMP	54 Timestamp request id=0x565d, seq=0/0, ttl=45
6	2.278236410	192.168.10.10	192.168.20.10	TCP	54 63903 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
7	2.283274947	192.168.10.10	192.168.20.10	TCP	58 63903 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
8	2.288583966	192.168.10.10	192.168.20.10	ICMP	42 Echo (ping) request id=0x0887, seq=0/0, ttl=41 (no response found!)
9	5.081729694	PCSSystemtec_5a:87:bc	PCSSystemtec_d1:26:3d	ARP	42 Who has 192.168.10.1? Tell 192.168.10.10
10	5.087706422	PCSSystemtec_d1:26:3d	PCSSystemtec_5a:87:bc	ARP	60 192.168.10.1 is at 08:00:27:d1:26:3d
11	12.522309779	192.168.10.10	192.168.1.1	ICMP	42 Echo (ping) request id=0x3154, seq=0/0, ttl=50 (reply in 21)
12	12.523321112	192.168.10.10	192.168.1.2	ICMP	42 Echo (ping) request id=0x024b, seq=0/0, ttl=46 (no response found!)
13	12.523693762	192.168.10.10	192.168.1.3	ICMP	42 Echo (ping) request id=0x4dd9, seq=0/0, ttl=56 (no response found!)
14	12.529932471	192.168.10.10	192.168.1.4	ICMP	42 Echo (ping) request id=0x56fb, seq=0/0, ttl=46 (no response found!)
15	12.530012526	192.168.10.10	192.168.1.5	ICMP	42 Echo (ping) request id=0x678c, seq=0/0, ttl=56 (no response found!)
16	12.530071023	192.168.10.10	192.168.1.6	ICMP	42 Echo (ping) request id=0x9087, seq=0/0, ttl=51 (no response found!)
17	12.530125448	192.168.10.10	192.168.1.7	ICMP	42 Echo (ping) request id=0x36a4, seq=0/0, ttl=45 (no response found!)



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fe80::7590:67b6:fa04:6…	ff02::2	ICMPv6	62	Router Solicitation
2	38.661961651	192.168.10.10	192.168.20.10	ICMP	42	Echo (ping) request id=0x910e, seq=0/0, ttl=45 (no response found!)
3	38.662129647	192.168.10.10	192.168.20.10	TCP	58	62438 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
4	38.662212973	192.168.10.10	192.168.20.10	TCP	54	62438 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
5	38.662333264	192.168.10.10	192.168.20.10	ICMP	54	Timestamp request id=0xc130, seq=0/0, ttl=37
6	40.667161531	192.168.10.10	192.168.20.10	ICMP	54	Timestamp request id=0x3447, seq=0/0, ttl=54
7	40.667975799	192.168.10.10	192.168.20.10	TCP	54	62440 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
8	40.668894470	192.168.10.10	192.168.20.10	TCP	58	62440 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
9	40.669536847	192.168.10.10	192.168.20.10	ICMP	42	Echo (ping) request id=0xce7b, seq=0/0, ttl=55 (no response found!)
10	43.966435727	PCSSystemtec_5a:87:bc	PCSSystemtec_d1:26:3d	ARP	42	Who has 192.168.10.1? Tell 192.168.10.10
11	43.973825183	PCSSystemtec_d1:26:3d	PCSSystemtec_5a:87:bc	ARP	60	192.168.10.1 is at 08:00:27:d1:26:3d

Breakdown of the logs

The image shows a Wireshark packet capture detailing network activity from a host at `192.168.10.10`.

The logs capture active network reconnaissance, including host discovery (ping scanning) and targeted communication attempts that are failing to receive responses.

🔍 Detailed Traffic Breakdown

1. Targeted Communication & Probing (Packets 1–8)

- **The Source:** A single host at `192.168.10.10`. (Kali Linux)
- **The Target:** A specific destination at `192.168.20.10` (Ubuntu server located on a different subnet).
- **The Activity:**

    - Sent ICMP Echo (ping) requests and ICMP Timestamp requests.
    - Attempted TCP connections to port 443 (HTTPS) via a `[SYN]` packet and port 80 (HTTP) via an `[ACK]` packet.
- **The Outcome:** Every single one of these packets is flagged with **`(no response found!)`**. The target is either offline, blocking the traffic with a firewall, or there is a routing issue preventing the return path.

2. Local Gateway Resolution (Packets 9–10)

- **The Activity:** An Address Resolution Protocol (ARP) exchange occurs on the local network.
- **The Details:** `192.168.10.10` broadcasts a request asking *"Who has 192.168.10.1?"* (likely its default gateway). The gateway responds, mapping its IP to the MAC address ending in `d1:26:3d`.
- **The Significance:** This confirms that local layer-2 and layer-3 communication on the source's immediate subnet is functioning perfectly.

Phase 2: Execution Commands (Examples)

If you need to re-run common activities from previous weeks, use these highly visible scanning commands:

- **Ping Sweep (ICMP Recon):**
  bash

```
None


nmap -sn 192.168.1.0/24
```

- Use code with caution.
- **Stealth TCP SYN Scan:**
  bash

None

```
sudo nmap -sS -F 192.168.20.X
```

- Use code with caution.
- **Aggressive Service/OS Detection:**
  bash

None

```
sudo nmap -A -T4 192.168.10.X
```

- Use code with caution.

Phase 3: Side-by-Side Traffic Comparison

Metric / Behavior	Pre-Firewall (Baseline)	Post-Firewall (Hardened)
Target Host Visibility	Hosts respond to ICMP pings and scan requests.	Hosts appear down or completely unresponsive (filtered).
Open Ports Discovered	Lists all active services (e.g., 22, 80, 443).	Shows `0 ports open` or all targeted ports as `filtered`.
Wireshark Traffic Pattern	Clear **SYN** → **SYN-ACK** → **RST** handshake sequences.	**SYN** packets sent by attacker, but **no response** (Timeouts) or **ICMP Unreachable** returned by the firewall.
Scan Completion Time	Fast (seconds to minutes) because the attacker gets immediate responses.	Significantly slower or times out because the attacker is waiting for dropped packets.

Firewall Logs	No alerts or drop actions logged.	Logs populating with `DROP`, `REJECT`, or `BLOCKED` flags from the attacker's IP.

Phase 4: Wireshark Analysis Filters

Use these display filters in Wireshark to quickly pinpoint what changed between your two capture files:

- `ip.src == [Attacker_IP]` → Isolates all traffic coming from your scanning machine.
- `tcp.flags.syn == 1 && tcp.flags.ak == 0` → Displays incoming connection attempts (port scans).
- `icmp` → Checks if ping sweeps are getting through or if the firewall is sending "Destination Unreachable" messages back.